EE2002-Tutorial 7 Solutions

1. Consider the circuit shown in Figure 1. We want this circuit to count down from 15 to 0. Suggest what changes are required and draw the new circuit. Also, explain the operation of your (new) circuit.

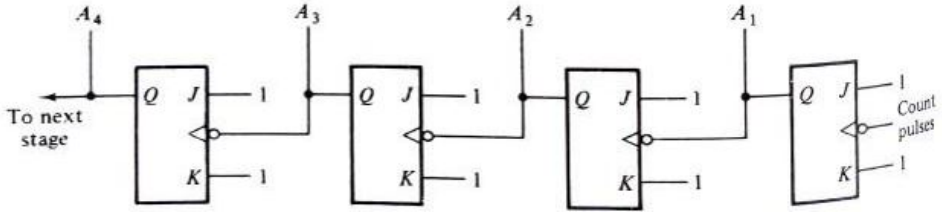


Figure 1: Ripple Counter

Solution:

A 4-bit ripple counter is a sequential circuit that counts in binary. The given circuit is an up-counter, and we need to modify it to count down from 15 to 0. The same circuit will work, we just need to take the output from Q' instead of Q. Figure 2 shows the solution.

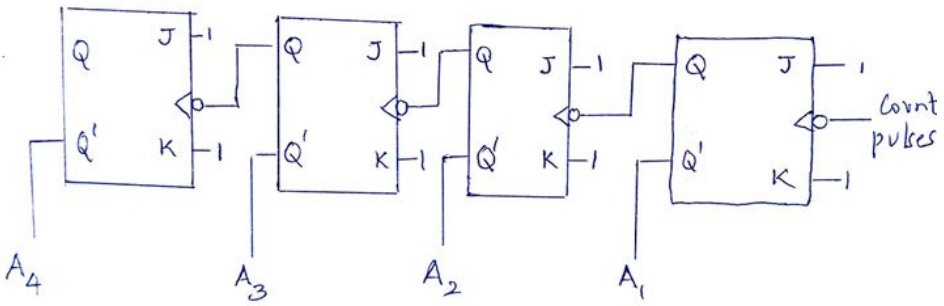


Figure 2: Ripple Counter

The down counter will count as follows.

Clock Cycle	A_4	A_3	A_2	A_1
0	1	1	1	1
1	1	1	1	0
2	1	1	0	1
3	1	1	0	0
4	1	0	1	1
5	1	0	1	0
6	1	0	0	1
7	1	0	0	0
8	0	1	1	1
9	0	1	1	0
10	0	1	0	1
11	0	1	0	0
12	0	0	1	1
13	0	0	1	0
14	0	0	0	1
15	0	0	0	0

2. An Up-Down counter circuit is shown in Figure 3. Study the behaviour of this circuit for all possible inputs (binary values of Up and Down).

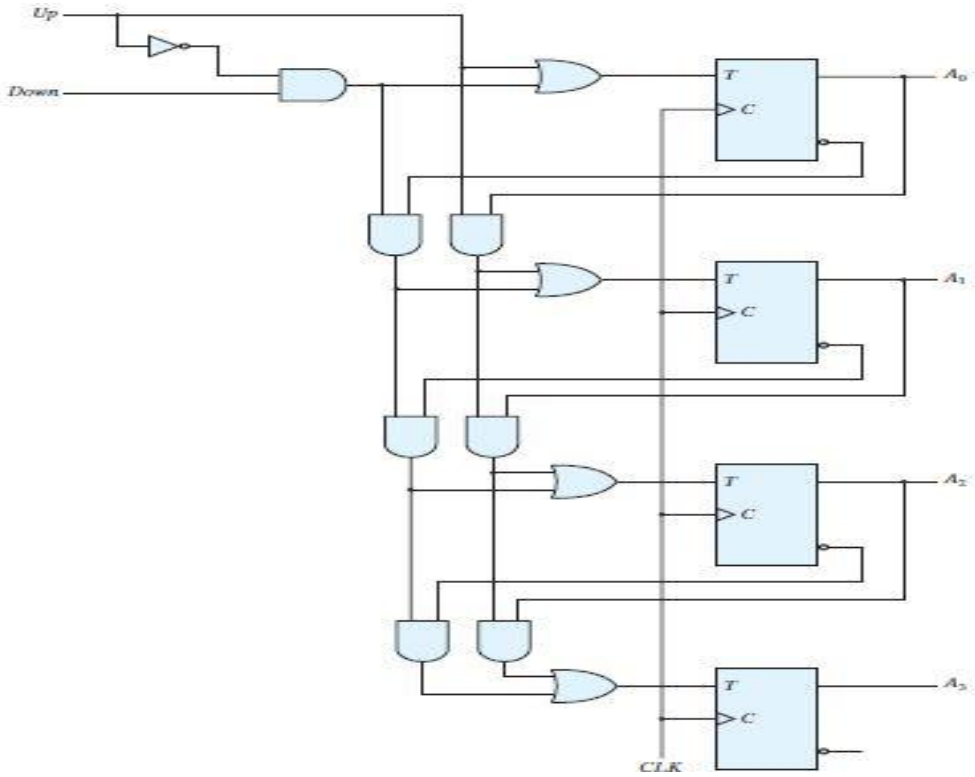


Figure 3: Up-Down Counter

Solution:

Circuit Behavior for Different Inputs

An Up-Down counter is a sequential circuit that can count in both increasing and decreasing order based

on control signals. The behavior of the counter is determined by two control inputs: **Up** and **Down**. **Behavior Analysis** The counter's behavior can be summarized as follows:

- If **Up** = 1 and **Down** = 0, the counter increments (counts up).
- If **Up** = 0 and **Down** = 1, the counter decrements (counts down).
- If **Up** = 0 and **Down** = 0, the counter holds its current state (no change).
- If **Up** = 1 and **Down** = 1, the counter behaves like an up counter and increments.

Explanation

Case 1: Up = 0, Down = 0 When both inputs are 0, there is no active triggering signal for the flip-flops. Since neither the normal nor the complemented outputs are being fed into the toggle (T) inputs, the circuit does not change state. It remains at its current count.

Case 2: Up = 0, Down = 1 In this case, the complemented outputs of the previous flip-flops are applied to the T inputs. This causes the counter to decrement, effectively counting down.

Case 3: Up = 1, Down = 0 Here, the normal outputs of the previous flip-flops are connected to the T inputs, making the counter increment. Thus, the counter counts up.

Case 4: Up = 1, Down = 1 Since both control signals are active, there is a conflict in logic. However, based on the circuit design, the Up signal dominates and the counter continues to increment (counts up). This is because the logic circuit prioritizes feeding the normal outputs of the flip-flops to the T inputs over their complemented outputs.

Truth Table

Up	Down	Operation
0	0	Hold (No Change)
0	1	Count Down
1	0	Count Up
1	1	Count Up

Conclusion The Up-Down counter functions based on the control inputs and either increments or decrements the count accordingly. If both inputs are 0, the counter remains in the same state, and if both inputs are 1, the counter continues counting up due to circuit prioritization.

3.Determine the operation of the circuit shown in Figure 4. Choose n = 3 and write down the sequence of outputs for 6 cycles. Now let n = 8. What is the (new) sequence ? Proceed until the sequence repeats.

Solution:

The given circuit is a Johnson counter, which is a type of shift register with feedback. It consists of n D flip-flops connected in series. The inverted output of the last flip-flop is fed back to the input of the first flip-flop.

The Johnson counter generates a sequence of states that is twice the number of flip-flops used ($2n$).

Sequence for $n = 3$

For $n = 3$, we have three D flip-flops, and the sequence will repeat every $2n = 6$ cycles. The table below shows the state transitions.

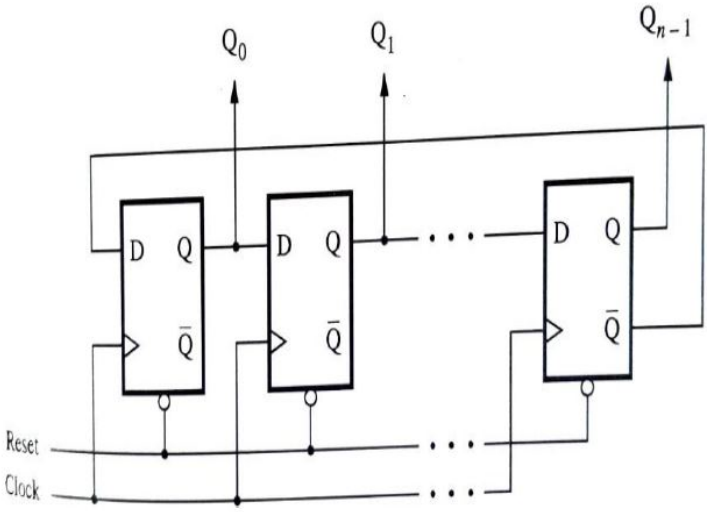


Figure 4: Another counter

Q_0	Q_1	Q_2
0	0	0
1	0	0
1	1	0
1	1	1
0	1	1
0	0	1
0	0	0

The sequence repeats after 6 cycles.

Sequence for $n = 8$

For $n = 8$, the sequence will repeat every $2n = 16$ cycles. The table below shows the state transitions.

Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7
0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0
1	1	1	0	0	0	0	0
1	1	1	1	0	0	0	0
1	1	1	1	1	0	0	0
1	1	1	1	1	1	0	0
1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1
0	0	1	1	1	1	1	1
0	0	0	1	1	1	1	1
0	0	0	0	1	1	1	1
0	0	0	0	0	1	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	0	1
0	0	0	0	0	0	0	0

The sequence repeats after 16 cycles.

4.What are the successive values of Q_0 and Q_1 in the circuit shown in Figure 5 ? Assume initial values for Q_1 and Q_0 to be 0 and 0 respectively.

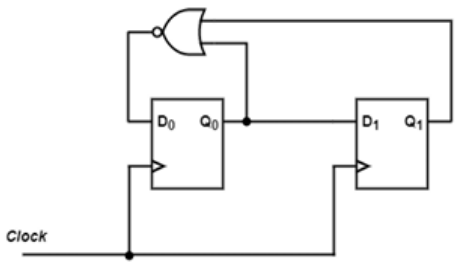


Figure 5: 2-bit Sequence

Solution:

From the given circuit:

$$D_0 = \overline{Q_0 + Q_1}$$
$$D_1 = Q_0$$

State Transition Table

Q_0	Q_1	$D_0 = \overline{Q_0 + Q_1}$	$D_1 = Q_0$	Q_0^+	Q_1^+
0	0	1	0	1	0
1	0	0	1	0	1
0	1	0	0	0	0
0	0	1	0	1	0

Counting Sequence

$\therefore$ The counting stages are $(00) \rightarrow (10) \rightarrow (01) \rightarrow (00) \rightarrow \ldots$

5.The circuit in Figure 6 operates like a counter. What is the counting sequence for this circuit ?

Assume the initial values as follows: $Q_0 = Q_1 = Q_2 = 0$**.**

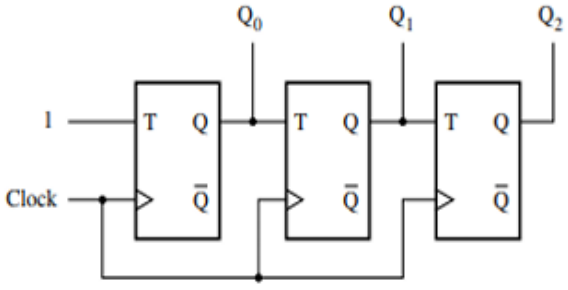


Figure 6: Three T Flip-Flops

Solution:

Initial States

$$Q_0 = 0, \quad Q_1 = 0, \quad Q_2 = 0$$

Clk	Q_0	Q_1	Q_2
0	0	0	0
1	1	0	0
2	0	1	0
3	1	1	1
4	0	0	0

Counting Sequence

$$\therefore \text{The counting stages are } (000) \rightarrow (100) \rightarrow (010) \rightarrow (111) \rightarrow \dots$$

6.Consider the sequential circuit shown in Figure 7. Find the maximum clock frequency (in MHz), at which the given circuit can operate reliably. Note that t_{pd}, t_{setup} and t_{hold} denote the propagation delay, setup time and hold time respectively.

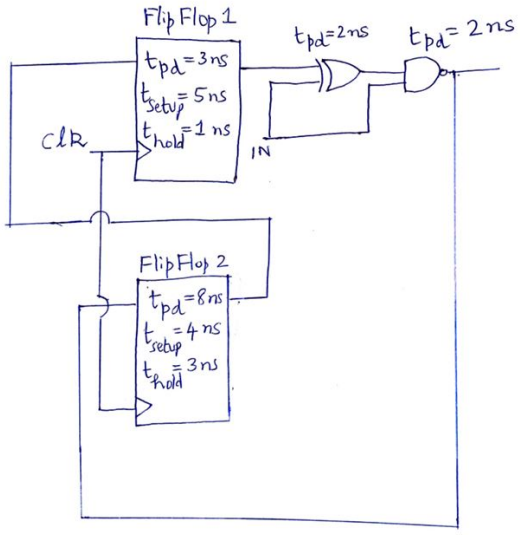


Figure 7: Circuit with specific values for propagation delay, setup time and hold time

Solution:

Setup Time: It is the amount of time for which the data at the synchronous input must be stable before the active edge of the clock.

Hold Time: It is the amount of time for which the data at the synchronous input must be stable after the active edge of the clock.

Figure 8 shows graphically how setup time and hold time are measured.

Since we have two flip-flops to which the clock is connected, we will have to calculate the minimum time period starting from both the flip-flops.

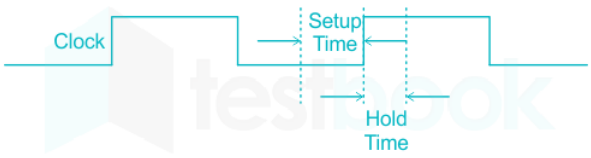


Figure 8: setup time and hold time

Starting from Flip-Flop 1: Output of flip-flop 1 is going as input to flip-flop 2 but through combinational logic. We can say that if the input to flip-flop 1 is stable at a (positive) clock edge, it will take t_{pd} (of flip-flop 1) i.e., 3 ns for its output to become stable. Thereafter, for the input of flip-flop 2 to become stable, it will take additional time determined by the sum of the propagation delay of the XOR gate (2 ns) and the propagation delay of NAND gate (2 ns). Note that the input to flip-flop 2 should still be stable at least for t_{setup} of flip-flop 2 (i.e., 4 ns) before the next (positive) clock edge arrives. So the total time period, t_p

$$t_p \geq (3 + 2 + 2 + 4)$$

Hence, the minimum time period = 11 ns. There is no role of hold time.

Starting from Flip flop 2: Similar arguments can be given. Since the output of FF2 is the input to FF1, we once again need to consider three components: propagation delay of FF2, propagation delay of combinational logic (if any) between FF2 and FF1 and setup time of FF1. In this case, there is no combinational logic between FF2 and FF1 so we only need to sum up the propagation delay of 8 ns (FF2) and setup time of 5 ns (FF1) to get 13 ns as the minimum period. Once again, there is no role of hold time.

The overall time period will be max (11, 13) since the "slower stage" (or the one that takes most time) will decide the reliability. The maximum frequency therefore is (1/13 ns) = 76.92 MHz.

7.Consider the synchronous sequential circuit in Figure 9. Analyse the circuit by writing the input equations.

Also, draw the state transition table.

	$\mathbf{Q_1}$	$\mathbf{Q_0}$
A :	0	0
B :	0	1
C :	1	1
D :	1	0

Solution:

From circuit Boolean Equations are:

$$J_1 = xQ_0$$

$$K_1 = x'$$

$$J_0 = x$$

$$K_0 = x + Q_1'$$

$$Z = xQ_1Q_0$$

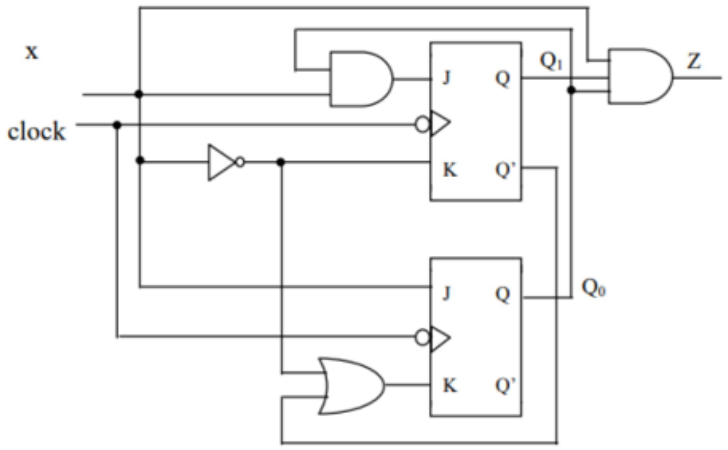


Figure 9: Synchronous Sequential Circuit

Q1Q0X	J1K1	J0K0	Q1+	Q0+	Z
000	0 1	0 1	0	0	0
001	0 0	1 1	0	1	0
010	0 1	0 1	0	0	0
011	1 0	1 1	1	0	0
100	0 1	0 0	0	0	0
101	0 0	1 1	1	1	1
110	0 1	0 0	0	1	1
111	1 0	1 1	1	0	0

Table 1: State transition table for the synchronous sequential circuit with JK flip-flop excitation and output.

8.A T flip-flop has a 15ns delay from the time its clock (pulse) input goes from 1 to 0 to the time the output is complemented. What is the maximum delay in a 8-bit binary ripple counter that uses this type of flip flop? What is the maximum frequency at which the counter can operate reliably ?

Solution:

The delay occurs when the output is complemented in the T flip-flop:

$$\text{Delay per TFF} = 15ns$$

The maximum delay in an 8-bit binary ripple counter using T flip-flops is:

$$\text{Maximum Delay} = 15ns \times 8 = 120ns$$

For the normal functioning of the counter, this delay should complete before the next clock edge arrives.

Therefore, the time period of the clock should be greater than the maximum delay:

$$T_{\text{clock}} \geq 120ns$$

The minimum time period of the clock is:

$$T_{\text{min}} = 120ns$$

The maximum frequency of operation is given by:

$$f_{\max} = \frac{1}{T_{\min}} = \frac{1}{120ns} = 8.33 \text{ MHz}$$

The maximum frequency at which the counter can reliably operate is **8.33** MHz.